Assignment 2

Solving Recurrences

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2.4.1 Using the substitution method, show that the solution of the recurrence $T(n) = T(n-1) + n$ is $O(n^2)$. Establish it also using recursion tree method.

Sol. **Substitution Method**

Assume **inductive hypothesis**: $T(k) \leq ck^2$ for some $c > 0$ for all $k < n$.

Then,

$$T(n) = T(n-1) + n$$

Step:

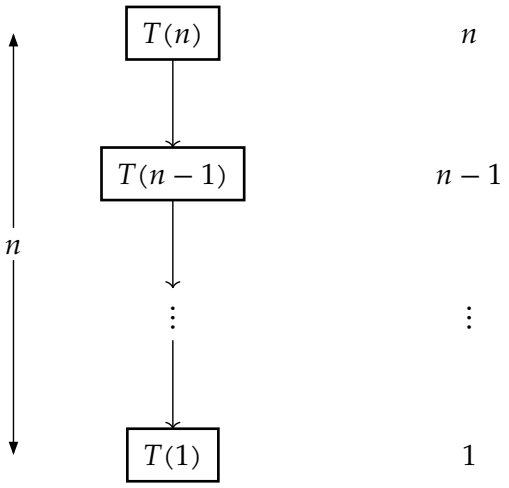
$$\begin{aligned} T(n) &\leq c(n-1)^2 + n \\ &= cn^2 - 2cn + c + n \\ &\leq cn^2 \quad \text{for } c \geq 1, n > 1 \end{aligned}$$

Suitable basis condition is $c = 1, n_0 = 2$

This proves that $T(n) = O(n^2)$.

Recursion Tree Method

$$T(n) = T(n-1) + n$$



At each level of the recursion tree, input size decreases by 1. Hence the number of levels is n. Each level has exactly one node and work done at kth node is $n - k$, so total time across each level is $\sum_{i=1}^{n} i = \frac{n(n+1)}{2} \Rightarrow T(n) = O(n^2)$.

2.4.2 Show that the solution of $T(n) = T(\lceil n/2 \rceil) + 1$ is $O(\lg n)$.

Sol.

Claim:

$$n_k = \underbrace{\lceil ... \lceil n/2 \rceil .../2 \rceil}_{k \text{ ceilings}} = \lceil n/2^k \rceil$$

Proof of claim: (Induction over k)

Basis: For $k = 1$, statement is trivially true. $\lceil n/2 \rceil = \lceil n/2^1 \rceil$.

Step: Assume that the statement is true for $k = m - 1$. Then $n_{m-1} = \underbrace{\lceil ... \lceil n/2 \rceil .../2 \rceil}_{m-1 \text{ ceilings}} = \lceil n/2^{m-1} \rceil$

For $k = m$,

$$n_m = \underbrace{\lceil ... \lceil n/2 \rceil .../2 \rceil}_{m \text{ ceilings}} = \lceil n_{m-1}/2 \rceil = \lceil \lceil n/2^{m-1} \rceil /2 \rceil$$

Let us take $2^{m-1} = p$. We need to simplify $\lceil \lceil n/p \rceil /2 \rceil$.

Let us write n in terms of $2p$. We can write $n = 2pq + r$, for $0 \leq r < 2p$, with unique $q, r \in \mathbb{Z}$.

Case 1: $r = 0$

$$\lceil \lceil n/p \rceil /2 \rceil = \lceil \lceil (2pq)/p \rceil /2 \rceil = \lceil \lceil 2q \rceil /2 \rceil = q$$

Case 2: $0 < r \leq p$

$$\lceil \lceil n/p \rceil /2 \rceil = \lceil \lceil (2pq + r)/p \rceil /2 \rceil = \lceil \lceil 2q + r/p \rceil /2 \rceil$$

Here $0 < \dfrac{r}{p} \leq 1$, so we get $\lceil (2q + 1)/2 \rceil = \lceil q + 1/2 \rceil = q + 1$.

Case 3: $p < r < 2p$

$$\lceil \lceil n/p \rceil /2 \rceil = \lceil \lceil (2pq + r)/p \rceil /2 \rceil = \lceil \lceil 2q + r/p \rceil /2 \rceil$$

Here $1 < \dfrac{r}{p} < 2$, so we get $\lceil (2q + 2)/2 \rceil = \lceil q + 1 \rceil = q + 1$.

Thus,

$$\lceil \lceil n/p \rceil /2 \rceil = \begin{cases} q & r = 0 \\ q + 1 & r \neq 0 \end{cases} \tag{1.1}$$

Notice that this is equivalent to the expression $\lceil n/2p \rceil$:

Case 1: $r = 0$

$$\lceil n/2p \rceil = \lceil 2pq/2p \rceil = q$$

Case 2: $0 < r \leq 2p$

$$\lceil n/2p \rceil = \lceil (2pq + r)/2p \rceil = \lceil q + r/2p \rceil$$

Here $0 < \dfrac{r}{2p} \leq 1$, so we get $q + 1$.

Thus we have

$$\lceil n/2p \rceil = \begin{cases} q & r = 0 \\ q + 1 & r \neq 0 \end{cases} \tag{1.2}$$

From (1.1) and (1.2) we can say

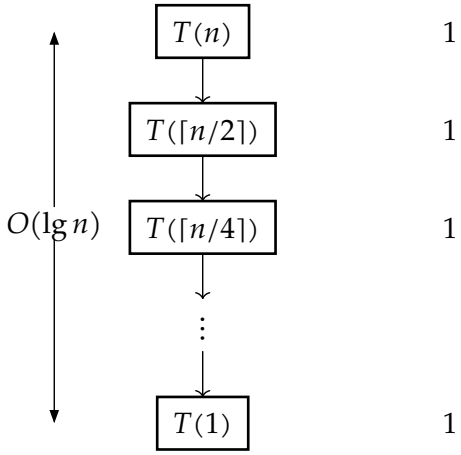
$$\lceil \lceil n/p \rceil /2 \rceil = \lceil n/2p \rceil$$

For $p = 2^{m-1}$, we have $\left\lceil \left\lceil n/2^{m-1} \right\rceil /2 \right\rceil = \lceil n/2^m \rceil$. This proves our inductive step and thus completes the induction. Thus our claim is verified.

Let's solve the given recurrence using **Recursion Tree Method**.

By using the **claim**, we can simplify the recurrence nodes. At kth level, the recursive step will be $T(\underbrace{\lceil \ldots \lceil n/2 \rceil \ldots /2 \rceil}_{k \text{ ceilings}}) = T(n_k) = T\left(\left\lceil n/2^k \right\rceil\right)$.

$$T(n) = T(\lceil n/2 \rceil) + 1$$



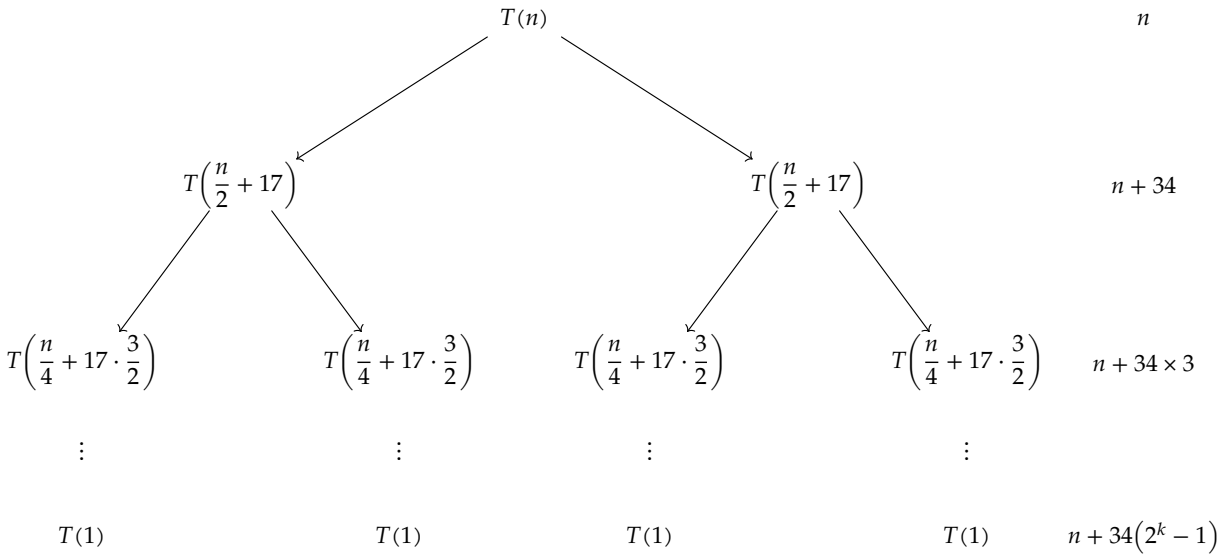
Each level has one node, and extra work at each level is constant. Thus, the height of the tree itself gives the total work, which is $O(\lg n)$.

$$T(n) = O(\lg n)$$

2.4.3 Using Recursion Tree Method, show that $T(n) = 2T(\lfloor n/2 \rfloor + 17) + n$ is $O(n \lg n)$.

Sol. Since we need to find the upper bound, we can approximate $\lfloor n/2 \rfloor$ as $\frac{n}{2}$, and this won't change the asymptotic behaviour of the recurrence.

$$T(n) = 2T\left(\frac{n}{2} + 17\right) + n$$



Work at kth level:

$$
\begin{aligned}
&= 2^k \times \left(\frac{n}{2^k} + 17 \times \left(1 + \frac{1}{2} + \ldots + \frac{1}{2^{k-1}}\right)\right) \\
&= n + 17 \times \left(2 + 4 + \ldots + 2^k\right) \\
&= n + 34 \times \left(1 + 2 + 4 + \ldots + 2^{k-1}\right) = n + 34 \times \left(2^k - 1\right)
\end{aligned}
$$

Total levels $= \Theta(\lg n)$.

Summing up all the works for each level, we have

$$
\begin{aligned}
T(n) &= \sum_{k=1}^{\lg n} n + \sum_{k=1}^{\lg n} 34\left(2^k - 1\right) \\
&= n \lg n + O\left(2^{\lg(n)}\right) \\
&= n \lg n + O(n) \\
\Rightarrow T(n) &= O(n \lg n)
\end{aligned}
$$

2.4.4 Show that a substitution proof for $T(n) = 4T\left(\frac{n}{3}\right) + n$ fails with $T(n) \leq cn^{\log_3 4}$, and how to fix it by subtracting a lower order term.

Sol.

$$T(n) = 4T\left(\frac{n}{3}\right) + n$$

Hypothesis $T(m) \leq cm^{\log_3 4}$ for all $m < n$.

Step

$$\begin{aligned} T(n) &= 4T\left(\frac{n}{3}\right) + n \\ &\leq 4c\left(\frac{n}{3}\right)^{\log_3 4} + n \\ &= cn^{\log_3 4} + n \end{aligned}$$

This way, we cannot show $T(n) \leq cn^{\log_3 4}$.

We can fix it by making a stronger inductive hypothesis, by subtracting a lower order term.

Hypothesis $T(m) \leq cm^{\log_3 4} - dm$ for all $m < n$.

Step

$$\begin{aligned} T(n) &= 4T\left(\frac{n}{3}\right) + n \\ &\leq 4c\left(\frac{n}{3}\right)^{\log_3 4} - 4d\left(\frac{n}{3}\right) + n \\ &= cn^{\log_3 4} - \frac{4}{3}dn + n \\ &= cn^{\log_3 4} - dn - n\left(\frac{1}{3}d - 1\right) \\ &\leq cn^{\log_3 4} - dn \qquad \text{for } d > 3 \end{aligned}$$

This completes the proof.

2.4.5 Solve the recurrence $T(n) = T\left(\sqrt{n}\right) + \lg n$ by making a change of variables. Give an asymptotically tight bound. Do not worry about whether values are integral.

Sol. Assume $n = 2^k$. Then we have

$$\begin{aligned} T\left(2^k\right) &= T\left(\sqrt{2^k}\right) + \lg 2^k \\ \Rightarrow T\left(2^k\right) &= T\left(2^{\frac{k}{2}}\right) + k \end{aligned}$$

Define a new function $S(k) = T\left(2^k\right)$.

$$S(k) = S\left(\frac{k}{2}\right) + k$$

Clearly, $S(k) \geq k$.

If we use recursion tree method, we find that

$$S(k) = k + \frac{k}{2} + \ldots + 1 \leq 2k$$

Together, this implies $S(k) = \Theta(k)$.

But $S(k) = T\left(2^k\right) = \Theta(k)$.

Put $k = \lg n$

$$\Rightarrow T(n) = \Theta(\lg n)$$

2.4.5 Solve the recurrence $T(n) = 2T\left(\sqrt{n}\right) + \lg n$ by making a change of variables. Give an asymptotically tight bound. Do not worry about whether values are integral.

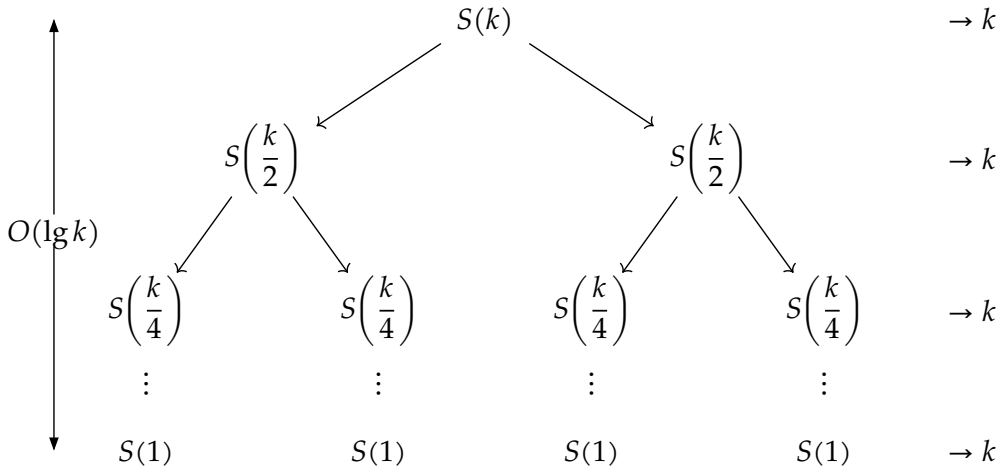
Sol. Assume $n = 2^k$. Then we have

$$T\left(2^k\right) = 2T\left(\sqrt{2^k}\right) + \lg 2^k$$

$$\Rightarrow T\left(2^k\right) = 2T\left(2^{\frac{k}{2}}\right) + k$$

Define a new function $S(k) = T\left(2^k\right)$.

$$S(k) = 2S\left(\frac{k}{2}\right) + k$$



By recursion tree, we can see that

$$S(k) = \Theta(k \lg k)$$

This is because, at each level total work will be $\Theta(k)$, and there will be $\lg k$ levels.

But $S(k) = T\left(2^k\right) = \Theta(k \lg k)$.

Put $k = \lg n$

$$\Rightarrow T(n) = \Theta(\lg n \cdot \lg \lg n)$$
